Introduction

America has a historic opportunity to build, secure, and control[3] a world-defining technology: artificial superintelligence (ASI). The right kind of superintelligence project could give the United States a decisive – possibly permanent – advantage over our adversaries, and deliver unprecedented economic benefits globally. But right now, we aren't building the right kind of superintelligence project. Research on the critical path to ASI is happening in badly secured private labs that are easy targets for CCP penetration, is not being strongly integrated with national security efforts, and is cutting corners on the critical technical problem of controlling these systems. All this can be fixed, and it will have to be.

From 2022 to early 2024, our team conducted an investigation of the national security implications of frontier AI development. We shared the first-ever publicly reported statements from frontier lab whistleblowers who were concerned that their labs' security posture was putting U.S. national security at risk. According to lab insiders, leading AI model weights – which are increasingly key U.S. national security assets – were being regularly stolen by nation-state actors. Critical lab IP was protected by flimsy security measures. And executives at some top labs were shutting down calls for heightened security coming from their own concerned researchers. In one case, a researcher told us about a running joke within their lab that they're "the leading Chinese AI lab because probably all of our [stuff] is being spied on."[4]

A few months later, OpenAI fired Leopold Aschenbrenner – a researcher who had written an internal memo criticizing what he considered OpenAI's poor security – allegedly for leaking sensitive information. [redacted]

[3] There are reasons to believe that artificial superintelligence — meaning AI systems that are significantly more capable than humans at all tasks — can't be controlled in any meaningful way. Obviously, whether that turns out to be true should be a huge factor in determining our national security strategy for ASI. We'll discuss this more later in the document.

[4] These concerns were justified. There have been many public reports since then of Chinese efforts – some successful – to spy on American hyperscalers and AI labs. For example, Chinese hacking group "Diplomatic Specter" has been spearphishing OpenAI employees; a former Google engineer stole AI-related trade secrets and siphoned them to Chinese firms; and Chinese state-backed hacking groups were able to obtain a cryptographic key from Microsoft that gave them full access to encrypted sensitive data from cloud servers. The scale of China's state-backed hacking program is "larger than that of every other major nation, combined", and has included infiltration of major U.S. broadband providers, and theft of private data used to train AI models.

According to one former researcher, "The government doesn't really know what's going on in the labs, and they're not even allowed to talk to people at the labs [...] Labs are trying to crack down on [technical researchers] that they suspect of potentially talking to the government."

As a result, the U.S. government lacks situational awareness regarding the capabilities of frontier models trained on American soil – increasingly critical national security assets. And unlike the U.S. government and its intelligence agencies, the CCP is free to spy on American companies: at any given time, the CCP may have a better idea of what OpenAI's frontier advances look like than the U.S. government does.

Most U.S. frontier labs have improved their security measures somewhat since we published our report. But as we'll see, these improvements are not enough to keep them ahead of current or future CCP operations.

Soon after his departure from OpenAI, Aschenbrenner published Situational Awareness, an influential manifesto that argued that the current explosive growth in AI capabilities, measured on a scale of "effective compute," suggests that superintelligence may very well be developed before the end of the decade.

AI scaling curves show that as AI models have been trained using ever larger quantities of compute and data, their capabilities have increased steadily and predictably across many orders of magnitude. As AI system training and inference is scaled on larger and larger superclusters, it's likely that the capabilities of frontier systems will continue to increase.

Those capability increases could happen surprisingly fast: AI is already augmenting important parts of the AI research process itself, and that will only accelerate.[5] Soon, AI research will be mostly or entirely automated, with millions of AI agents pushing the field forward at computer speeds rather than human speeds, and getting faster with every exponential increase in computing power. On the way there, we might get warning shots – if we're lucky, shocking demos; if we're unlucky, high-consequence attacks or accidents – that will fully awaken national security agencies to the stakes of superintelligence. At that point, Aschenbrenner argued,

[5] Frontier lab insiders we spoke to have told us that as of February 2025, as much as 50% of their lab's codebase is already written by AI. They expect this number to reach 75% by the end of 2025.

everything will change. Proposals that seem outrageous today will slide across the Resolute Desk — from locking down the labs to outright nationalization of ASI research.

Leopold isn't alone in predicting that we're on course for superintelligence, or that more government involvement in AI is inevitable. Our team has spent the last four years working closely with a network of insiders at top AI labs and U.S. national security agencies to understand what the march to ASI will actually look like — and what it will call upon us to do, if we want the U.S. to come out ahead of the CCP, and the cause of liberty to win through. In that time, the views of many of our contacts in the AI labs have sharpened: as we approach superintelligence, it's become clearer to them that AI is turning into a national security technology first and foremost. To quote a former cyber intelligence operator we interviewed during our investigation, "Dual-use is a nice way to [describe superintelligence], but actually it'll be a weapon for all intents and purposes."

A small number of people in the frontier AI world saw this coming, and they weren't just employees at leading labs. As early as 2020, a technical background and some discussions with the right friends at OpenAI were enough to convince us to dive head-first into AI national security. Even then, it was clear that we'd end up with an industry-wide race to build AI infrastructure, and that the national security implications of AI development were soon going to be impossible to ignore.

To this day, if you know the right people, the Silicon Valley gossip mill is a surprisingly reliable source of information if you want to anticipate the next beat in frontier AI – and that's a problem. You can't have your most critical national security technology built in labs that are almost certainly CCP-penetrated, some of which have a history of shutting down whistleblowers with security concerns. As a former DIU executive told us, "if legislators knew how compromised [redacted AI lab] might be, there would be a significant appetite to shut this all down."[6] Despite deepening partnerships with national security agencies, frontier AI development is leaky: most lab executives aren't pricing in how effective nation-state

[6] He didn't mean to shut down the frontier AI labs, but rather to move the development of high-stakes AI R&D out of highly penetrated companies with huge security vulnerabilities.

espionage can be.[7] Shoring up lab security to the point where we aren't giving away the crown jewels of American technological superiority to the CCP will require fundamental changes to the culture, management, and operations of frontier AI labs.

Right now, the greatest danger is not that the U.S. will fall behind China in the race to superintelligence. Until we've secured the labs, *there is no lead for us to lose*. Just the opposite: as we've seen, U.S. national security agencies don't constitutionally spy on American companies or access their technology illicitly, but the CCP has no such scruples. Under the status quo, therefore, advances at private U.S. labs may lead to advances in CCP capabilities *before* they lead to advances in U.S. national security capabilities.[8]

Apart from security, there's also the question of what chain of command should apply to future superintelligent systems. We're already seeing AI models with meaningful bioweapon design, cyber, autonomy, and even persuasion capabilities. Superintelligence will go beyond that: it will be *the* national security technology. It will deliver a decisive, and possibly permanent strategic advantage to whoever builds and controls it first – and it may not be easy to control. It must be built by, secured in, and controlled from America. And it must be operated in a way that serves the interests of the American people.

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[7] One former OpenAI researcher pushed back on this framing as applied to OpenAI, and we're including his perspective here for completeness even though it's not a universally held view. As he put it, "It's not that [OpenAI] aren't pricing this in. It's that they don't care. They know they are going to get away with it; it's not like they are going to get fired for having bad security, whereas if they were to beef up their security and thereby fall behind in the race to AGI, they would lose all their power and chance at glory etc. My guess at least, is that they cynically calculate that the U.S. government will at most slap them on the wrist for having bad security. tbc this is just my guess, I don't have direct knowledge of what they are thinking. But they aren't idiots, they know the CCP has probably penetrated them and could easily do so in the future."

[8] In fact, according to national security professionals we interviewed – many of whom had experience in the IC – under today's conditions, a very real possibility is that 1) A U.S. frontier lab gets noticeably close to building superintelligence; 2) the CCP observes this progress and exfiltrates what it needs to reproduce it domestically; 3) the CCP begins to sabotage the lab's further development efforts, likely in an obfuscated or deniable way; and 4) the CCP hands over the stolen technology to its national champions to finish the work themselves. In other words, while some U.S. frontier labs believe internally that they can achieve superintelligence first by pushing AI capabilities as fast as possible, that belief is quite possibly deluded — in reality, they're more likely to be beaten at the last minute by unscrupulous actors with access to nation-state resources.

If we end up launching a national project, the need for that project will be recognized abruptly, and when it is, we'll wish we'd had more time to think through how to pull it off. For the past year, we've been working to answer exactly that question: how might a U.S. government-backed superintelligence project work? We interviewed dozens of frontier AI researchers, energy specialists, AI hardware engineers, national security executives and action officers, and constitutional lawyers. We visited hyperscaler data centers with former Tier 1 special forces operators to assess the state of the art of data center security. We spoke to dozens of Congressional offices to understand where the Overton window is today, and how far it will have to move to catch up to the decade of superintelligence.

This might seem premature. But if progress simply continues on its current path, we could suddenly find ourselves in a technological and geostrategic crisis – maybe even a hot war – over AI. And if that happens, we'll wish we'd had more time[9] to anticipate contingencies, figure out how to design a CCP-proof project, find out whether it can do what we want it to do – and decide how to run it.

Scope of the investigation

AI is already a dual-use technology. But as we get closer to superintelligence, it will be seen more and more as an enabler and driver of weapon of mass destruction (WMD) capabilities, if not as a WMD in and of itself. Direct calls for a "Manhattan Project for AGI" are already starting.

But it's still not obvious how exactly the U.S. government could or should partner with frontier AI labs on a superintelligence program. Should frontier labs be nationalized outright?[10] Should they be put on contract to support U.S. government research activities? Or should the

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[9] Many of the security measures we'll need to pull off a successful superintelligence project – especially those that require technical R&D – will take months to years to implement.

[10] As part of the research that informed our March 2024 report, we spoke to frontier lab insiders and whistleblowers, many of whom were concerned that their labs were brushing off their increasingly critical security concerns. During a lab tour, a researcher discreetly took Jeremie aside. Referring to our eventual recommendations to the U.S. government, he told him that, "I would urge you to be more ambitious," and asked us to recommend that frontier AI research be nationalized. That would have been an extreme measure, particularly at the time. But he wasn't the only one calling for it even then, almost a year before the Situational Awareness memo brought the idea into the mainstream national security conversation.

relationship between the government and the labs be at arm's length, with the government offering incentives – such as funding or infrastructure – to shape frontier AI research?

There's already been some informed speculation about the answers to these questions, but our goal here isn't to get bogged down in debates about the specific configuration of a government-backed superintelligence project. Instead, we're going to focus on identifying the implementation-level problems that a public-private partnership on superintelligence would have to solve in order for it to achieve critical U.S. national security objectives.

There are many ways that a government-backed ASI project could be set up, but few that will lead to a lasting American advantage without inadvertently handing our most important national security technologies to our adversaries. We'll be going after a very small target while wielding a very large industrial and national security machine.

We'll start by introducing some of the main considerations that our investigation has surfaced, sketching out a path to a U.S.-backed superintelligence program.